



INVENTORY APP FOR DISASTER RELIEF

31-10-2025 | Follow-Up Meeting

Content

01



Project scope

Our Project Understanding +
Main Functionalities

02



App Design

Initial Drafts

03



Architecture

Technologies, Limits +
Opportunities

04



Organization

Teams + Milestones

Project Scope ✕ *Minimum Viable Product (Local)*



Excel Import/ Export



Interactive Data Base

→ *Add Items*

→ *Search and Locate Items*

→ *Track Availability and Condition*



Maintenance & Inspection



Vehicle Packing Plan

Project Scope ✕ Backlog



QR Code Scanning



Configurable User Notifications



Cloud Integration:

→ *Data Encryption*

App Design ✕



(Interactive) Table as Core Functionality



Design clarity: buttons, icons, layout



Responsiveness

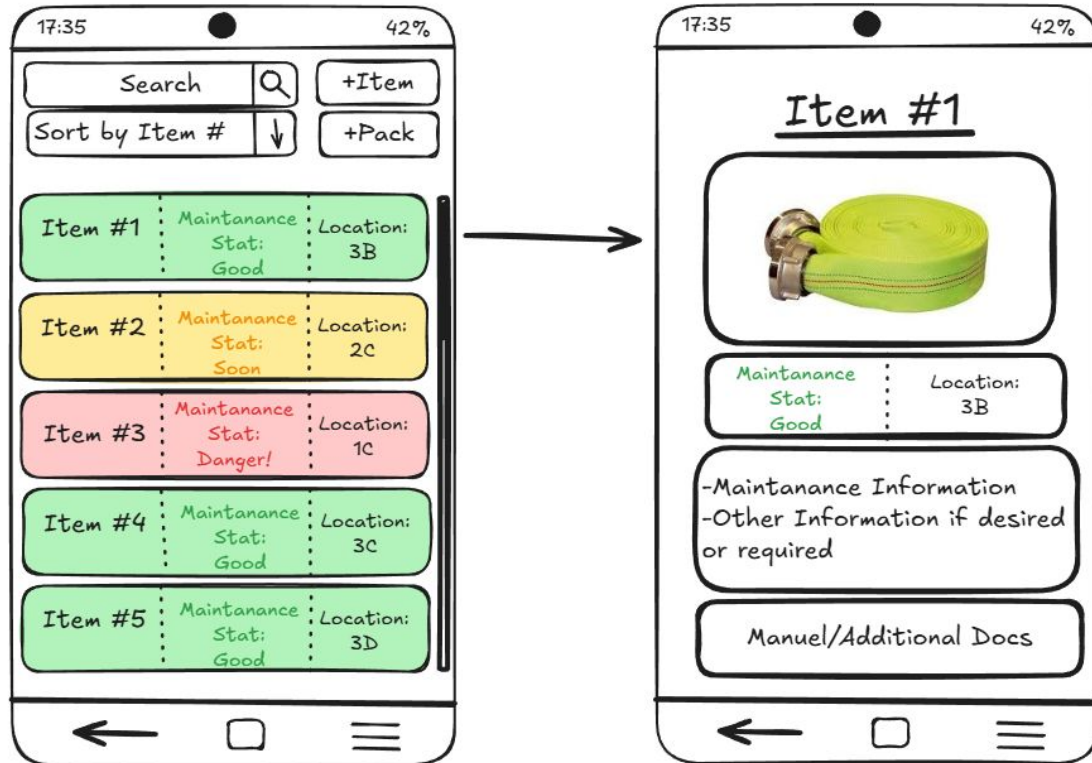


Clear feedback and guidance

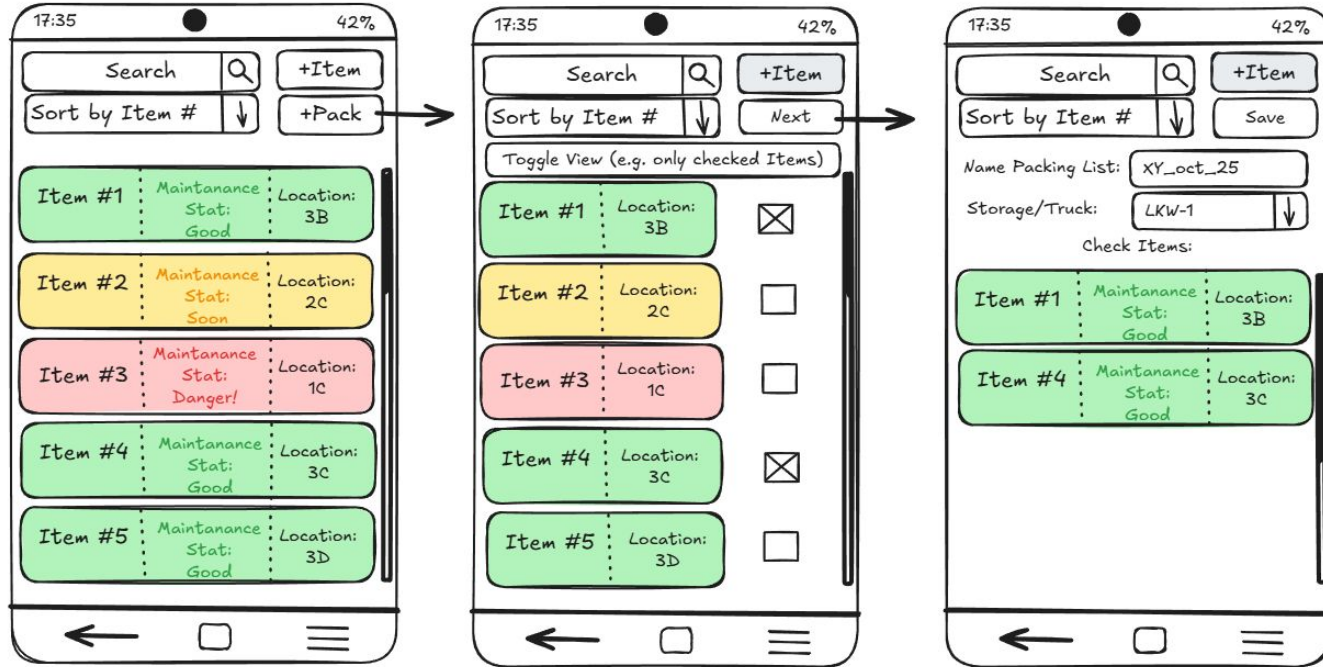
UI Design ✕ Main Menu



UI Design ✕ Mobile Inventory



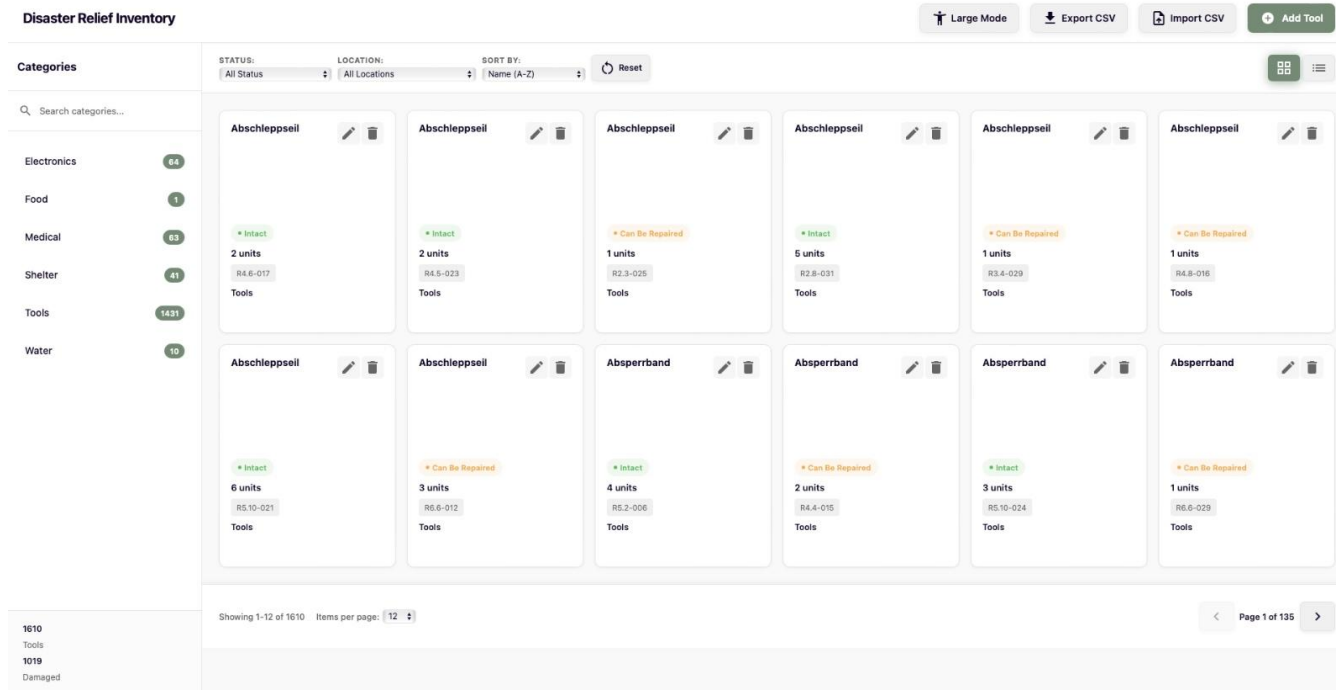
UI Design ✕ Packing Plan



UI Design ✖ List-Based Inventory (Desktop)

Disaster Relief Inventory				<button>Large Mode</button>		<button>Export CSV</button>	<button>Import CSV</button>	<button>Add Tool</button>
Categories Search categories... Electronics 84 Food 1 Medical 83 Shelter 41 Tools 1431 Water 10	STATUS:		LOCATION:	SORT BY:				
	All Status		All Locations	Name (A-Z)	<button>Reset</button>			
	Abschleppseil	Intact	2 units	R4.8-017	Tools			
	Abschleppseil	Intact	2 units	R4.9-023	Tools			
	Abschleppseil	Can Be Repaired	1 units	R2.3-025	Tools			
	Abschleppseil	Intact	5 units	R2.8-031	Tools			
	Abschleppseil	Can Be Repaired	1 units	R3.4-029	Tools			
	Abschleppseil	Can Be Repaired	1 units	R4.9-015	Tools			
	Abschleppseil	Intact	6 units	R5.10-021	Tools			
	Abschleppseil	Can Be Repaired	3 units	R6.8-012	Tools			
1610 Tools 1019 Damaged	Absperrband	Intact	4 units	R5.2-009	Tools			
	Absperrband	Can Be Repaired	2 units	R4.4-015	Tools			
	Absperrband	Intact	3 units	R5.10-024	Tools			
	Absperrband	Can Be Repaired	1 units	R6.5-029	Tools			

UI Design ✕ Tile-Based Inventory (Desktop)



UI Design ✕ Add Tool (Desktop)

Add New Tool

MANUAL ENTRY

SCAN QR CODE

TOOL NAME *

e.g., First Aid Kit

QUANTITY *

0

LOCATION *

e.g., Warehouse A, Shelf 3

STATUS *

✓ Select Status

Intact

Can Be Repaired

Permanently Damaged

CATEGORY *

Select Category

NEXT MAINTENANCE (OPTIONAL)

30/10/2025

Cancel

Add Tool

Architecture ✕ In General



*(separately) maintainable division into “Frontend”,
“Backend”, “Data Base”*



PWA over Native Apps



highest device compatibility; “deploy once, run anywhere”



offline storage; later-on: synchronizable with external storage



*limits negligible due to simple scope
→ ~500MB storage limit; no bluetooth/ USB peripherals; some
iOS hassles*

Architecture ✕ Technologies



Vite.js + Typescript:



high-performing development server and build tool



ensure code quality, improve developer experience; crucial for interdisciplinary team



IndexedDB



*offline storage; later: local copy of cloud data
(necessary for when there is no service in disaster zones)*



high-capacity



no maintenance needed; always-on

Architecture ✕ *Local Data Storage*



no dedicated Backend



mock API to lay groundwork for cloud integration;

-->React.js-based

Organization ✕



Communication: *WhatsApp, Discord*



Documentation: *In-Code, Readme (RWTH-GitLab, technical perspective),
User Manual (PDF; user perspective)*



Responsibilities

DATA MANAGEMENT

Adrien

Aziz

DESIGN + FRONTEND

Imane

Julian

LOGIC + BACKEND

Ewa

Julian

Focus on Interdisciplinary Teamwork; everyone participates; everyone's opinion matters

Organization ✕ Milestones



[M1] Interactive Database

Midterm



"as-is" functionality provided by Excel but smarter, user-friendlier



Import/ Export



Tool Locating



[M2] Maintenance & Inspection Planning

Final



automated maintenance reminders

Organization ✕ Milestones



[M3] Vehicle Packing Plan

Final



user-assisted packing of tools for various disaster scenarios



Presets?



[M4] Tool Tracking



QR-Codes/ Barcode Scanning

Organization ✕ *Bonus Milestones*



[B1] Cloud Architecture Preparation



containerized database + backend



[B2] Cloud Integration



data encryption



server deployment

PROJECT TIMELINE



QUESTIONS ✕

- Cloud Integration; especially: working with multiple devices
 - export whole dataset and send to other devices
- Table Dummy Data:
 - UUIDs?
 - Sample values for empty columns
 - How many locations/rooms are there?
 - Size of the table: 2x dummy dataset
- Design Preferences; Mobile-First => List instead of Tiles
- Grading Scheme (=> Scalability, UI Design, Met Aspirations, ...)
 - Core Application, Documentation, Self-Organization (Soft Skills)
- GitLab-Access: User Permissions
- Documentation: PDF needed? Technical documentation/reasoning

- Ebene: **ignore**
- Ort (how to destructure components)
 - R6.7-006
 - R: Rolling Container; G: Box, size of Euro-Palette
 - R6 = 6th Rolling Container
 - .7 = Box inside container
 - -006 = Number of Tool inside the Box
- Ausstattung | Hersteller | Typ
 - probably: any of the things
- Sachnummer: meaning?
 - vs. Inventar-Nr. (UID, fixed format, 4 digits-6digits), vs. Gerätenr.
- FB: **ignore**
- Status: intact, defective + description field; some things do not need maintenance (e.g. screwdrivers)

Feedback

- Tool for Creating QR Code would make sense
- User Feedback should be gathered continuously
 - Mariia and Malte
- Upload slides to GitLab
- Initial-Setup:
 - *How-To* should be included in each feature (e.g. packing plan)
 - Functionality: Item View should include “Quick Action”: *Mark as Used, Mark as Available, ...*
 - Documentation: GitLab-Wiki instead of Readme
 - Communication: Mariia does not need to supervise communication, but welcomes additional communication channels with Malte and her (Discord, WhatsApp)
- User Manual makes sense; definitely needs good developer documentation; report of what has been done in the end